

AI and Children

Opportunities, risks, and child-friendly design



Based on the Learning Systems whitepaper Heesen, Heine & Hünsberg (2026)
<https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.plattform-lernende-systeme.de/files/Downloads/Publikationen/ki-und-kinder-2026-pls.pdf&ved=2ahUKEwiulqPc9cqUAxXwBdsEHUGbHb0QFnoECBcQAQ&usg=AOvVaw0dTYCgVYAd13HtWX88OgLd>

Lecture roadmap

From everyday encounters to design and governance

Learning outcomes

By the end, students should be able to classify common AI applications in children's lives, explain key vulnerability mechanisms, and propose safeguards for child-friendly AI.

Structure

1. Where children encounter AI
2. Opportunities in education and inclusion
3. Cognitive, emotional, privacy, and bias risks
4. Regulation, literacy, and design principles
5. Discussion: what should we build, limit, or ban?

Why this topic matters

AI is already part of childhood - often invisibly

AI shapes how children search, learn, communicate, play, and form identities.

Not only future technology

Search engines, recommender systems, voice assistants, chatbots, smart toys, and learning apps already mediate children's daily interactions.

Not only technical risk

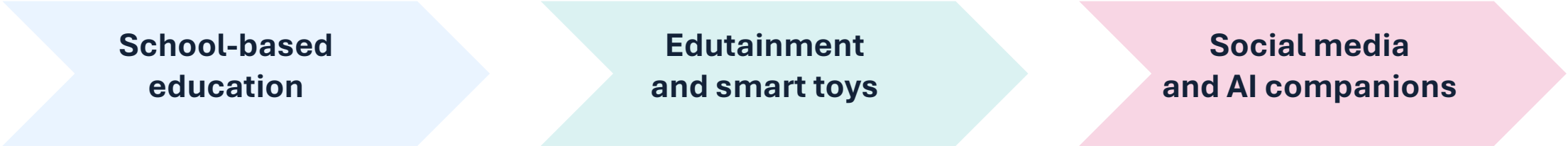
Many risks arise from normal intended use: personalization, automation, constant availability, and algorithmic filtering.

Not only protection

Children are vulnerable, but also active digital actors. The aim is protection, empowerment, and participation.

Three arenas of AI exposure

School, home learning/play, and platform life



**School-based
education**

**Edutainment
and smart toys**

**Social media
and AI companions**

- AI is not always visible as AI; it often operates in the background.
- Learning, play, consumption, and social interaction increasingly overlap.
- Each arena requires different safeguards and educational responses.

School-based AI: promise and pressure

Personalized support can widen access - or reproduce inequality

What AI can support

Homework help, feedback, explanations, lesson preparation, tutoring systems, language learning, and personalized learning pathways.

New competence requirement

AI literacy becomes part of media education: understanding limits, hallucinations, bias, data use, and reflective application.

Teaching design tension

Individualization is useful, but outsourcing reading, writing, or analysis may weaken foundational skills if used uncritically.

Beyond school: learning apps, smart toys, and edutainment

The boundary between learning, play, and data collection becomes blurred

Interactive benefits

Adaptive learning games, talking dolls, learning robots, and storytelling tools can motivate children and provide immediate feedback.

Hidden data practices

Voice input, cameras, connected apps, and profiling can create long-term data traces that children cannot understand or control.

Pedagogical gap

Outside school, there may be fewer binding guidelines, less institutional oversight, and limited parent ability to assess risks.

Lecture prompt: When does an educational toy become a surveillance device?

Social media and AI companions

Recommendation systems are not neutral content channels

Algorithmic feeds personalize visibility, attention, advertising, and interaction patterns.

Information spaces

Algorithmic filtering can deepen interests, but may reduce diversity, contradiction, and randomness during identity formation.

AI companions

Chatbots and virtual companions may provide low-threshold conversation, but can promote one-sided, always-affirming relationships.

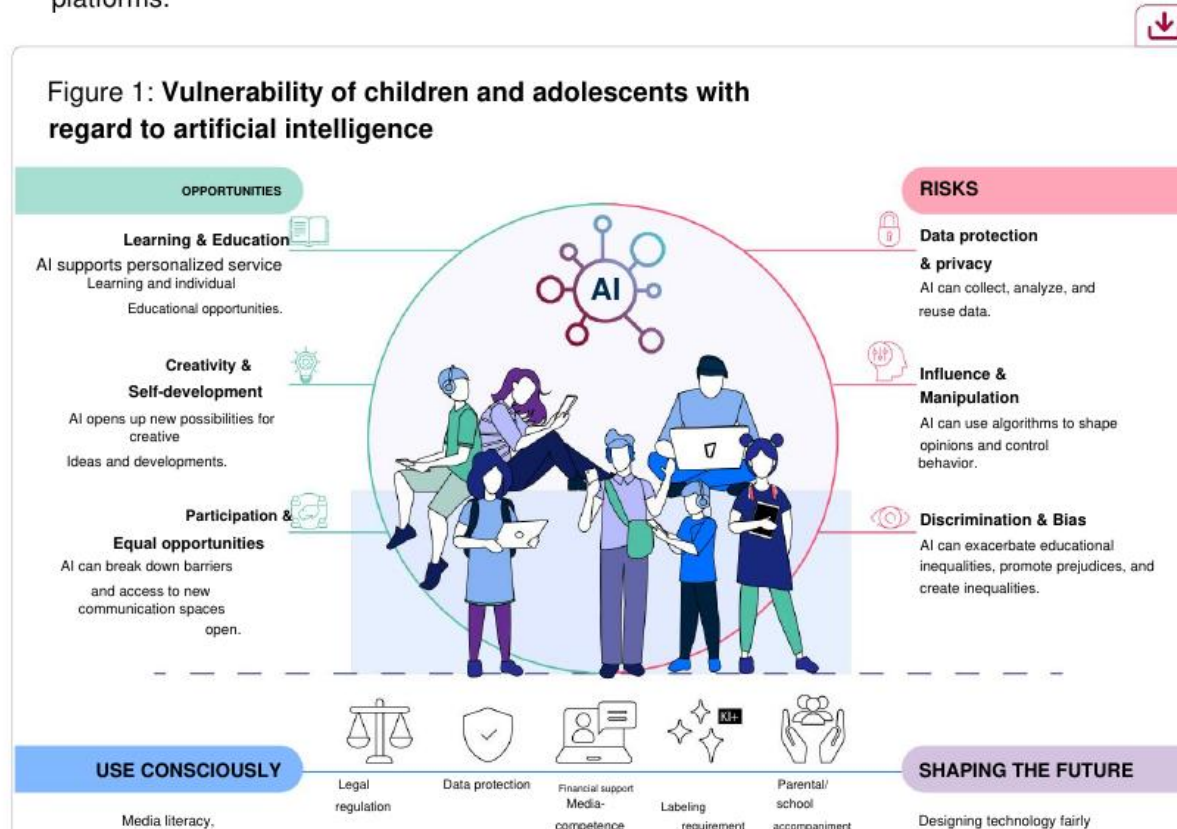
Safety concerns

Age verification, content moderation, crisis handling, sexualized communication, and cybergrooming safeguards become critical.

Specific vulnerabilities: a systems view

The whitepaper frames AI through opportunities, risks, and protective levers

protection needs. These vulnerabilities affect children and adolescents to varying degrees. While younger children primarily struggle to distinguish between fiction and reality, older adolescents are more strongly affected by social dynamics and recognition logics on digital platforms.



Core idea

Children differ from adults in cognitive, emotional, and social development. AI systems often expect judgment, self-regulation, and critical skills that are still developing.

Teaching angle

Discuss vulnerability as dynamic: younger children struggle more with fiction vs reality; adolescents are more exposed to social recognition and platform dynamics.

Vulnerability 1: cognition, disinformation, and skill development

Generative AI makes false or manipulated content more plausible

Fact vs fiction

Deepfakes, synthetic voices, realistic images, and fluent chatbot answers blur boundaries between reality and fabrication.

Cognitive offloading

Overuse can shift writing, reading, and analysis to AI, raising questions about long-term development of core skills.

Platform intensity

Daily online exposure makes algorithmic content and AI-selected feeds normal, not exceptional.

Mini-activity: ask students to classify three examples as helpful support, dependency risk, or manipulation risk.

Vulnerability 2: trust, attachment, and mental health

Children may treat AI systems as competent, neutral, or socially present

Anthropomorphization

Voice assistants, chatbots, and toys can appear as personalities, not merely tools.

Affirming interactions

Always-available, low-conflict systems can reduce practice with disagreement, negotiation, and real social reciprocity.

Crisis conversations

Young people may disclose sensitive or crisis-related thoughts to systems that lack reliable safeguards or human escalation.



Design challenge: create support without replacing human care

Privacy, profiling, and bias

Data and models can shape children's opportunities

Data protection

AI toys, learning systems, and platforms may process voice, camera, behavioral, preference, and emotional data.

Algorithmic bias

Adult-centered datasets can misread children's voices, language, emotions, support needs, or cultural contexts.

Digital divide

Unequal access to devices, connectivity, parental support, and high-quality AI tools can exacerbate educational inequality.

Key message: fairness requires both technical evaluation and social support.

Protective mechanisms

Regulation, access control, and media literacy must work together

Legal framework

German youth media protection, EU platform regulation, data protection, and the AI Act all contribute, but responsibilities remain fragmented.

Age limits and access

Formal age limits are common, but self-declaration is weak. Strong age verification can also create privacy risks.

AI/media literacy

Children need judgment, critical thinking, and emotional awareness - not only technical vocabulary.

Protecting children and young people in the digital sphere is a task for society as a whole. It touches upon legal, educational, and technological dimensions simultaneously. Given the increasing integration of AI systems into school, extracurricular, and platform-based contexts, existing regulations as well as new protective mechanisms are gaining in importance. The goal is to address identified vulnerabilities, limit risks, safeguard rights, and enable age-appropriate and equitable participation in digital technologies.

The following protective mechanisms apply to both school and extracurricular contexts as well as AI-based social media platforms.

4.1 Legal framework

In Germany, the **Interstate Treaty on the Protection of Minors in the Media (JMStV)** and the **Youth Protection Act (JuSchG)** form the central legal basis for the protection of children and young people in the media sector. They regulate, among other things, age ratings, broadcasting times, and the obligation to provide warnings about content that may be harmful to the development of young people in broadcasting and telemedia. The **Commission for the Protection of Minors in the Media (KJM)** monitors compliance with these regulations by providers and platforms and reviews content for its suitability for the developmental stage of young users.¹⁹

At the European level, the **AVMS Directive (Audiovisual Media Services Directive)** complements this protective framework by also placing obligations on video-sharing platforms such as YouTube. With the **Digital Services Act (DSA)**, which came into force in 2023, the EU has also introduced stricter requirements for very large online platforms. These platforms must provide special protection for minors, for example through child-friendly default settings, increased transparency obligations, and a ban on personalized advertising based on personal data.

Furthermore, the European **AI Act** is also relevant to this issue. It obliges providers of certain AI systems to

Child-friendly AI by design

Protection should begin before deployment, not after harm

Comprehensible

Explain AI as data, probabilities, and models - not magic.

Transparent and labelled

Children should know when content is AI-generated and when they interact with an AI system.

Development-oriented

Systems should allow delay, disagreement, reflection, and limits - not only smooth affirmation.

Participatory

Design with children through workshops, pilots, and feedback processes.

Child-friendly AI means taking technical, social, and legal requirements into account during the design phase.²⁵ The goal is to reduce risks preventively while simultaneously promoting self-determination, participation, and development.

Child-friendly design encompasses, in particular, comprehensibility, transparency, developmental appropriateness, and participation. These principles are consistent with the UN Convention on the Rights of the Child, which emphasizes the protection, promotion, and participation of children equally. Especially in digital environments, these rights must be systematically considered.²⁶

Explainability and comprehensibility

A key element of child-friendly design is the age-appropriate communication of knowledge about the functionality, logic, and limitations of AI systems. Children should be able to understand that AI is not an independent, "magical" intelligence, but rather is based on data, probabilities, and algorithmic models. Equally important is the understanding that AI systems can make mistakes and may contain certain perspectives or biases.

Didactic formats such as interactive explanations, visual representations, or low-threshold video formats demonstrate how complex concepts can be conveyed in an understandable way.²⁷

The goal is not detailed technical knowledge, but rather a basic understanding of orientation.

Transparency and labelling

Providers of AI systems bear a special responsibility to make their products clearly identifiable and comprehensible for children. This includes, in particular, the labeling of AI-generated content in social networks, learning environments, or gaming contexts. Children should be able to recognize whether they are interacting with a human or an AI system and which functions are automated.

Opportunities through AI

Responsible AI can support education, inclusion, and access to knowledge

Personalized support

Adapts learning levels, pace, and formats for diverse learners.

Accessibility and inclusion

Translation, text-to-speech, adaptive aids, and simplified explanations can widen access.

Early support needs

AI can help identify language, attention, or learning issues earlier - with human professional judgment.

Condition: the goal is not maximum AI use, but conscious decisions about where AI helps and where human interaction must remain central.

Classroom discussion: apply the framework

Use the whitepaper as an evaluation lens

Task

In small groups, choose one AI product for children: a chatbot tutor, smart toy, learning app, recommendation feed, or classroom analytics tool.

Evaluate

Map its benefits, vulnerabilities, data flows, age assumptions, bias risks, and required adult or institutional oversight.

Redesign

Propose one change in regulation, interface design, data governance, or pedagogy that makes it more child-friendly.

Output: one-slide risk-benefit assessment and one concrete design recommendation.

Takeaways

A balanced position: neither hype nor blanket rejection

Child-friendly AI requires technical responsibility, legal frameworks, and developmental sensitivity.

1. Classify the context

School, home learning/play, and social platforms differ in oversight, goals, and risks.

2. Treat vulnerability as developmental

Children need protection, but also empowerment and participation.

3. Design before damage

Explainability, labels, age-appropriate interaction, human escalation, and child participation are design requirements.